

SAT Math Mastery · Week 5A

Exponentials & Sequences

Time 150 minutes: 110 min solving + 40 min self-check and error log

Topics exponential growth and decay models · base and initial value · doubling and half-life · compound growth · exponent laws · arithmetic vs geometric sequences · linear vs exponential · tables and graphs

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Build $y = ab^t$ or $y = a(1 \pm r)^{t/k}$ from a verbal description (percent $\rightarrow$ factor; “every k units” $\rightarrow$ exponent t/k).
- S2.** Read the initial value and the growth or decay factor from an equation, a table or a graph.
- S3.** Use doubling time and half-life; solve for a time by matching powers of the same base.
- S4.** Tell exponential from linear: constant *ratio* versus constant *difference*.
- S5.** Apply the exponent laws, rational exponents and equal-base equations; rewrite equivalent exponential forms.
- S6.** Arithmetic sequences $a_n = a_1 + (n-1)d$ and geometric sequences $a_n = a_1r^{n-1}$ (including negative ratios).
- S7.** Interpret constants and factors in context; identify the growth type from “equals $p\%$ of the previous value”.
- S8.** Build a and b from two points; use parameters, shifts and hidden conditions.

2 · REQUIRED FACTS

- Exponential: equal steps in x multiply y by the same factor. Linear: equal steps in x add the same amount to y .
- $y = ab^x$ with $a > 0$: $b > 1$ is growth, $0 < b < 1$ is decay; a is the value at $x = 0$ and the graph never touches the x -axis.
- A change of $p\%$ per period has factor $1 + \frac{p}{100}$ (growth) or $1 - \frac{p}{100}$ (decay), never $\frac{p}{100}$ alone.
- “Every k units” puts $\frac{t}{k}$ in the exponent; the factor for one unit is the k -th root of the factor for k units.
- The exponent in a geometric sequence is $n - 1$, not n . Terms m apart differ by the factor r^m (or by md in an arithmetic sequence).
- A compounded balance over n periods is $P(1 + \frac{r}{n})^{nt}$ for annual rate r , n periods per year.

3 · REQUIRED FORMULAS

$$y = ab^x, \quad y = a(1 \pm r)^t, \quad y = ab^{t/k}, \quad a_n = a_1r^{n-1}, \quad a_n = a_1 + (n-1)d$$

$$b^m b^n = b^{m+n}, \quad \frac{b^m}{b^n} = b^{m-n}, \quad (b^m)^n = b^{mn}, \quad b^{-n} = \frac{1}{b^n}, \quad b^{m/n} = \sqrt[n]{b^m}, \quad b^0 = 1$$

$$\text{two points: } b^{x_2-x_1} = \frac{y_2}{y_1}, \quad A = P \left(1 + \frac{r}{n}\right)^{nt}, \quad \text{half-life: } N = N_0 \left(\frac{1}{2}\right)^{t/h}$$

4 · MINI THEORY SHEET

Linear or exponential?

	Linear	Exponential
Table	equal Δy for equal Δx	equal y_2/y_1 for equal Δx
Words	“increases by c each period”	“increases by $p\%$ each period”
Equation	$y = mx + b$	$y = ab^x$
Long run	grows by the same amount	eventually beats every linear model

Reading $y = ab^t$. a = value when $t = 0$. b = factor per unit of t ; the percent change is $(b - 1) \cdot 100\%$. If the exponent is t/k , then b is the factor per k units.

Rewriting the base. $b^{t/k} = (b^{1/k})^t$. Write both sides over one common base (powers of 2, 3, 5) and compare exponents.

From a table or two points. Divide the outputs and take a root of the step count: if x moves by m , then $b^m = \frac{y_2}{y_1}$. Then $a = y_1 b^{-x_1}$.

Sequences. Same ratio $\Rightarrow$ geometric, same difference $\Rightarrow$ arithmetic. Three consecutive terms p, q, r of a geometric sequence satisfy $q^2 = pr$; of an arithmetic sequence, $2q = p + r$.

Context. The constant out front is the starting amount; the base is the multiplier per period; never read the exponent as the percent.

Speed shortcuts.

Situation	Shortcut
doubling / halving	count how many doublings: 2^n ; target = start $\times 2^n$
percent $\rightarrow$ factor	$+7\% \rightarrow 1.07$, $-12\% \rightarrow 0.88$; multiply by the factor n times
every k units	exponent $\frac{t}{k}$; rewrite as $(b^{1/k})^t$ for a per-unit factor
solve for an expression	factor out 3^x (e.g. $3^{x+2} - 3^x = 8 \cdot 3^x$) and read off the block
two table rows	$b^{\Delta x} = \frac{y_2}{y_1}$, then work back to a with one equation
geometric three-in-a-row	$q^2 = pr$ gives the unknown immediately
linear vs exponential race	test values; the exponential only wins after a crossover month
shift in a graph	$f(x - h) = a b^{-h} b^x$: the shift changes only the leading constant

5 · GUIDED EXAMPLES

Example 1 — Build the model from words

A medicine concentration starts at 240 mg/L and falls by 30% every 4 hours. Write $C(t)$, t in hours, and find $C(8)$.

Factor per 4 hours: $1 - 0.30 = 0.70$. So $C(t) = 240(0.7)^{t/4}$.

$C(8) = 240(0.7)^2 = 240(0.49) = 117.6$ mg/L.

Example 2 — Two points $\rightarrow a$ and b

An exponential function has $f(1) = 12$ and $f(4) = 96$. Find $f(x)$ and $f(0)$.

Three steps from $x = 1$ to $x = 4$: $b^3 = \frac{96}{12} = 8 \Rightarrow b = 2$.

$12 = a(2)^1 \Rightarrow a = 6$. So $f(x) = 6(2)^x$ and $f(0) = 6$.

Example 3 — Match the base

Solve $4^{x+1} = 32^{x-2}$.

$4 = 2^2$ and $32 = 2^5$: $2^{2(x+1)} = 2^{5(x-2)}$. Exponents: $2x + 2 = 5x - 10 \Rightarrow x = 4$.

6 · PRACTICE SET — 48 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1.

EASY · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 3 \cdot 2^x$. What is the value of $f(5) - f(3)$?

2.

EASY · NO CALCULATOR NEEDED

The table shows values of the function f for selected values of x . Which of the following defines f ?

x	0	1	2	3
$f(x)$	64	48	36	27

(A) $f(x) = 64 \left(\frac{1}{4}\right)^x$

(B) $f(x) = 64 \left(\frac{3}{4}\right)^x$

(C) $f(x) = 64 - 16x$

(D) $f(x) = 64 \left(\frac{4}{3}\right)^x$

3.

EASY · NO CALCULATOR NEEDED

The balance, in dollars, of a savings account t years after the account is opened is modeled by $A(t) = 800(1.05)^t$. Which of the following statements best describes the account?

(A) Starts at \$800; up 105% per year.

(B) Starts at \$1.05; up \$800 per year.

(C) Starts at \$800; up 5% per year.

(D) Starts at \$800; up \$5 per year.

4.

EASY · NO CALCULATOR NEEDED

The first term of a geometric sequence is 6 and the common ratio is $\frac{1}{2}$. What is the fourth term of the sequence?

(A) $\frac{3}{4}$

(B) $\frac{3}{8}$

(C) $\frac{3}{2}$

(D) $\frac{3}{16}$

5.

EASY · NO CALCULATOR NEEDED

Which of the following is equivalent to $\frac{(2x^3)^3}{4x^2}$, for $x \neq 0$?

(A) $\frac{x^7}{2}$

(B) $2x^{11}$

(C) $2x^{\frac{9}{2}}$

(D) $2x^7$

6.

MEDIUM · NO CALCULATOR NEEDED

A lab culture contains 500 bacteria at time $t = 0$, where t is measured in hours, and the number of bacteria doubles every 3 hours. Which of the following equations gives the number of bacteria N at time t ?

(A) $N = 500(2)^t$

(B) $N = 500(2)^{3t}$

(C) $N = 500(2)^{t/3}$

(D) $N = 500(3)^t$

7.

MEDIUM · NO CALCULATOR NEEDED

A 640-milligram sample of a radioactive substance loses half of its mass every 5 days. After how many days will exactly 20 milligrams of the sample remain?

8.

MEDIUM · CALCULATOR

A delivery van is bought for \$28,000 and loses 12% of its value each year. To the nearest dollar, what is the value of the van 3 years after it was purchased?

(A) \$17,920

(B) \$19,081

(C) \$24,640

(D) \$39,338

9.

MEDIUM · NO CALCULATOR NEEDED

The exponential function f satisfies $f(2) = 18$ and $f(5) = 486$. What is the value of $f(0)$?

10.

MEDIUM · NO CALCULATOR NEEDED

The first term of a geometric sequence is 5 and the fourth term is 135. What is the second term?

- (A) 27 (B) 45
(C) 70 (D) 15
-
-
-
-
-
-
-

11.

MEDIUM · NO CALCULATOR NEEDED

Each year, the population of a town is 94% of its population the previous year. Which of the following types of function best models the town's population as a function of the number of years?

- (A) Exponential decay (B) Increasing linear
(C) Decreasing linear (D) Exponential growth
-
-
-
-
-
-
-

12.

MEDIUM · NO CALCULATOR NEEDED

The value, in dollars, of a collectible t years after 2020 is modeled by $V(t) = 1800(0.92)^t$. Which of the following is the best interpretation of the number 0.92 in this context?

- (A) The value in 2020 was \$0.92. (B) The value decreases by 8% each year.
(C) The value decreases by 92% each year. (D) The value decreases by \$0.92 each year.
-
-
-
-
-
-
-

13.

MEDIUM · NO CALCULATOR NEEDED

If $27^{x-1} = 9^{x+2}$, what is the value of x ?

14.

HARD · CALCULATOR

A savings account is modeled by $A(t) = 5000(1.015)^{4t}$, where $A(t)$ is the balance in dollars t years after the initial deposit. Which of the following is closest to the percent by which the balance increases each year?

(A) 1.50%

(B) 6.00%

(C) 6.14%

(D) 15.0%

15.

HARD · NO CALCULATOR NEEDED

The function f is defined by $f(t) = 90 \cdot 8^{t/6}$. Which of the following functions is equivalent to f ?

(A) $90(\sqrt{2})^t$ (B) $90 \cdot 2^{t/6}$ (C) $90\left(\frac{4}{3}\right)^t$ (D) $90 \cdot 4^{t/6}$

16.

HARD · CALCULATOR

For the exponential function $f(x) = a \cdot b^x$, where a and b are positive constants, $f(1) = 24$ and $f(4) = 81$. What is the value of $f(6)$?

(A) 243

(B) $\frac{2187}{8}$ (C) $\frac{243}{2}$ (D) $\frac{729}{4}$

17.

HARD · DESMOS

Channel A has 50 subscribers and gains 30 subscribers each month. Channel B has 50 subscribers and its subscriber count increases by 20% each month. What is the least whole number of months after which Channel B has more subscribers than Channel A?

18.

HARD · NO CALCULATOR NEEDED

The second term of a geometric sequence is 6, the fourth term is 54, and the common ratio is negative. What is the fifth term?

19.

HARD · CALCULATOR

A radioactive sample loses 36% of its mass every 5 days. Its mass can also be modeled by $M(t) = M_0 b^t$, where t is the number of days and b is a constant. What is the value of b , rounded to the nearest thousandth?

20.

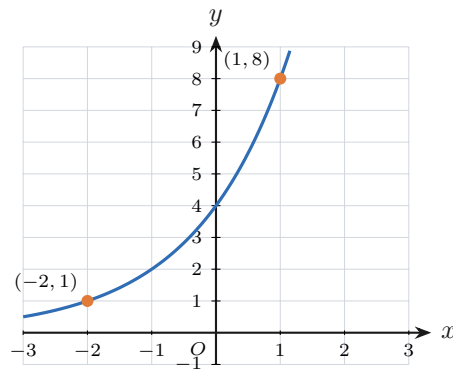
HARD · NO CALCULATOR NEEDED

If $3^{x+2} - 3^x = 216$, what is the value of 3^{x+1} ?

21.

HARD · DESMOS

The graph of the function $f(x) = a \cdot b^x$, where a and b are positive constants, is shown in the xy -plane. The points $(-2, 1)$ and $(1, 8)$ lie on the graph. What is the value of $a + b$?



(A) 10

(B) 12

(C) 6

(D) 8

22.

HARD · NO CALCULATOR NEEDED

The numbers $x + 2$, $x + 8$ and $x + 26$ are, in that order, three consecutive terms of a geometric sequence. What is the value of x ?

(A) 6

(B) 1

(C) 2

(D) 4

23.

HARD · NO CALCULATOR NEEDED

For positive x and y , the expression $\frac{(x^{2/3}y^{-1})^3}{x^{-1}y^2}$ is equivalent to x^py^q , where p and q are constants. What is the value of $p + q$?

(A) 0

(B) 8

(C) -8 (D) -2 **24.**

SAT CHALLENGE · NO CALCULATOR NEEDED

An arithmetic sequence a_n and a geometric sequence g_n both have first term 2. The common difference of a_n and the common ratio of g_n are both equal to k , where $k > 1$. If $g_3 - a_3 = 38$, what is the value of g_5 ?

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

For the function $f(x) = a \cdot b^x$, where a and b are constants and $b > 0$, $f(2) = 12$ and $f(4) - f(2) = 96$. What is the value of $f(5)$?

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

Culture A starts with 4000 cells and its population is cut in half every 2 hours. Culture B starts with 500 cells and its population doubles every 3 hours. After how many hours, t , will the two cultures have the same number of cells?

(A) 3.6

(B) 18

(C) 0.6

(D) 1.8

27.

SAT CHALLENGE · NO CALCULATOR NEEDED

The terms of a geometric sequence are $a_1, a_2, a_3, \dots$. The sum of the second and third terms is 12, and the sum of the third and fourth terms is 36. What is the value of a_6 ?

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

A quantity decreases by 19% every 2 years. Its current value is 4000. In how many years from now will its value be 2916?

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined by $f(x) = a \cdot b^x$, where a and b are positive constants. If $f(3) > f(2)$, which of the following statements must be true?

I. $f(-1) < f(0)$ II. $f(5) - f(4) > f(4) - f(3)$ III. $f(0) > 1$

(A) I only

(B) I and II only

(C) II and III only

(D) I, II and III

30.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 48 \left(\frac{2}{3}\right)^x$. The function g is defined by $g(x) = f(2x - 4)$ and can be written in the form $g(x) = c \cdot d^x$, where c and d are constants. What is the value of $c \cdot d$?

(A) 243

(B) 48

(C) 108

(D) 162

31.

SAT CHALLENGE · CALCULATOR

Real numbers x and y satisfy

$$\sqrt[3]{2^x} \cdot 4^y = 256 \quad \text{and} \quad \frac{16^x}{\sqrt[3]{4^y}} = 2^{22}.$$

What is the value of xy ?

32.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a real number x , $4^x - 2^{x+2} - 32 = 0$. What is the value of 2^{x+1} ?

(A) 16

(B) 32

(C) -8

(D) 8

33.

SAT CHALLENGE · CALCULATOR

The size of a bacterial colony, in thousands of cells, is modeled by $V(t) = 500 \cdot 2^{t/2} \cdot 4^{t/3} \cdot 16^{-t/12}$, where t is the number of hours since the start of observation. The size of the colony is multiplied by a constant factor c every 6 hours. What is the value of c ?

34.

SAT CHALLENGE · CALCULATOR

The function f is defined by $f(x) = \frac{9^x \cdot 12^{x+1}}{8^x}$. Which of the following expressions is equivalent to $f(x)$ and displays the value of $f(2)$ as a coefficient?

(A) $2187 \left(\frac{27}{2}\right)^{x+2}$

(B) $162 \left(\frac{27}{2}\right)^{x-1}$

(C) $\frac{729}{4} \left(\frac{27}{2}\right)^{x-2}$

(D) $2187 \left(\frac{27}{2}\right)^{x-2}$

35.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 2^x + k$, where k is a constant. For a real number p , $f(p) = 9$ and $f(-p) = \frac{21}{4}$. What is the value of p ?

36.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined by $f(x) = c \cdot b^{x+3}$, where c and b are positive constants. If $f(1) = 1296$ and $f(3) = 729$, then for each increase of 1 in the value of x , $f(x)$ decreases by $p\%$. What is the value of p ?

37.

SAT CHALLENGE · CALCULATOR

The value V , in dollars, of a delivery van x years after it is purchased is modeled by $V(x) = 18,000(0.49)^{x/2}$. Which of the following best describes how the van's value changes each year?

- (A) It decreases by 51% each year. (B) It decreases by 24.5% each year.
(C) It decreases by 30% each year. (D) It decreases by 70% each year.

38.

SAT CHALLENGE · CALCULATOR

The area covered by a coral colony increases by 33.1% every 3 years. If the colony grows by the same percent each year, by what percent does its area increase each year?

SAT CHALLENGE · CALCULATOR

SAT CHALLENGE • NO CALCULATOR NEEDED

$$\frac{6^{2x+3}}{2^{2x-1}} = k \cdot 9^x$$

41.

SAT CHALLENGE · CALCULATOR

A patient takes a 100-milligram dose of a medication at 8:00 a.m. and again at 8:00 p.m. every day, beginning at 8:00 a.m. on Monday. The dose is absorbed instantly, and between any two consecutive doses exactly half of the medication present in the body is eliminated. To the nearest tenth of a milligram, how much medication is in the body immediately after the 8:00 a.m. dose on Thursday?

- (A) 196.9 (B) 198.4
(C) 199.2 (D) 187.5

42.

SAT CHALLENGE · CALCULATOR

The function $P(t) = \frac{120}{1 + 5(0.8)^t}$ models the number of active users of an app, in thousands, t weeks after launch, for $0 \leq t \leq 10$. Consider the following statements.

- I. P is increasing on the interval $0 \leq t \leq 10$.
II. There is a time t in the interval at which $P(t) = 60$.
III. There is a time t in the interval at which $P(t) = 80$.

Which of the statements must be true?

- (A) I only (B) I and II only
(C) II and III only (D) I, II, and III

43.

SAT CHALLENGE · CALCULATOR

The function f is defined by $f(x) = a \cdot b^x$, where a and b are positive constants. For every value of x , $f(x + 2)$ is 44% greater than $f(x)$. For every value of x , $f(x + 3)$ is $p\%$ greater than $f(x)$. What is the value of p ?

(A) 132

(B) 60

(C) 66

(D) 72.8

44.

SAT CHALLENGE · NO CALCULATOR NEEDED

The value V , in dollars, of a piece of lab equipment t years after it was purchased is modeled by

$$V(t) = 2400(0.64)^{t/2.5}.$$

Which of the following is the best interpretation of the model?

(A) It falls 36% every 2.5 years.

(B) It falls 36% every 5 years.

(C) It falls 64% every 2.5 years.

(D) It falls 2.5% every year.

45.

SAT CHALLENGE · CALCULATOR

For all real numbers x ,

$$8^x \cdot 4^{x/3} \cdot (\sqrt{2})^x = b^x,$$

where b is a positive constant. Which of the following is equal to 2?

(A) $b^{-6/25}$

(B) $b^{6/25}$

(C) $b^{25/6}$

(D) b^{25}

46.

SAT CHALLENGE · NO CALCULATOR NEEDED

The function f is defined for $x \geq 0$ by

$$f(x) = 48 \left(\frac{1}{8} \right)^{\frac{2-x}{3}}.$$

Which of the following is equivalent to $f(x)$ for all $x \geq 0$ and displays the value of $f(0)$ as a constant factor?

(A) $f(x) = 12 \cdot \left(\frac{1}{2} \right)^x$

(B) $f(x) = 192 \cdot \left(\frac{1}{2} \right)^x$

(C) $f(x) = 12 \cdot 2^x$

(D) $f(x) = 48 \cdot 2^x$

47.

SAT CHALLENGE · NO CALCULATOR NEEDED

The number of bacteria in a colony t years after January 1, 2020 is modeled by $A(t) = 300 \cdot 4^{t/3}$. Which of the following gives the number of bacteria m months after January 1, 2023?

(A) $1200 \cdot 2^{m/18}$

(B) $300 \cdot 2^{m/18}$

(C) $1200 \cdot 2^{m/36}$

(D) $900 \cdot 2^{m/18}$

48.

SAT CHALLENGE · NO CALCULATOR NEEDED

Let $g(x) = 7 + 9\left(\frac{1}{4}\right)^x$ for all real numbers x . Consider the following statements.

I. $g(x) > 7$ for every real number x .

II. The minimum value of g is 7.

III. g is a decreasing function.

Which of the statements must be true?

(A) I and III only

(B) I and II only

(C) II and III only

(D) I, II, and III